

WASP-43b

R-0625

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180127 · created 2026-07-25T13:33:48+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458145.8147 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.171 +/- 0.012
Transit depth (Rp/Rs)^2	2.91 +/- 0.42 [%]
Orbital Inclination (inc)	82.52 +/- 1.29
Airmass coefficient 1 (a1)	0.984 +/- 0.011
Airmass coefficient 2 (a2)	0.0103 +/- 0.0045
Scatter in the residuals of the lightcurve fit is	1.1 %
Best Comparison Star	#3 - [343, 52]
Optimal Aperture	4.2
Optimal Annulus	21.58
Transit Duration (day)	0.0527 +/- 0.0038

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.171 $\pm$ 0.012 vs 0.1594 (deviation 0.96 σ)
Duration fit vs published	0.0527 d vs 0.0512 d (deviation 0.38 σ)
Mid-time O–C	-0.1 min
Depth SNR	14.1
Residual scatter	1.1 %

Light curve

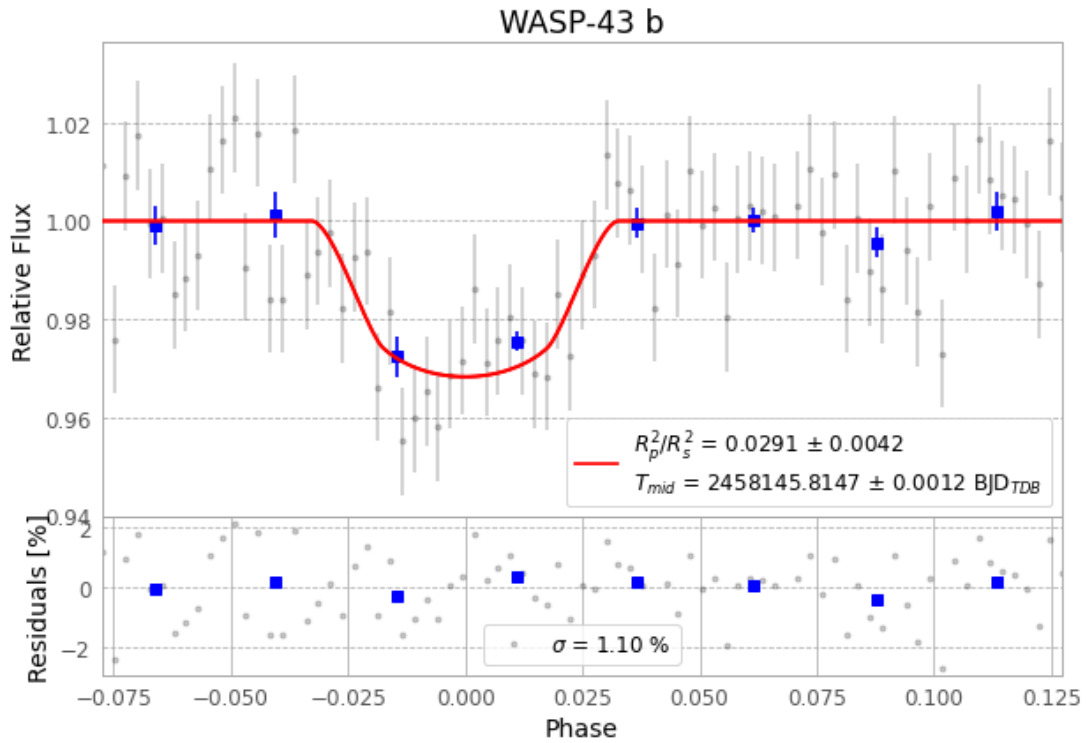


Figure 1 — FinalLightCurve_WASP-43 b_2018-01-26.png (SHA-256 de72327753c2254f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [430, 222], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5db4b1064df8b8d1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

81 FITS frames (53 MB), WASP-43180127055412.FITS → WASP-43180127095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180127.log (8006f2f4e819...), FinalLightCurve_WASP-43 b_2018-01-26.png (de72327753c2...), FinalLightCurve_WASP-43 b_2018-01-26.csv (5497641052ec...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 13:33Z.